

# Midterm Exam

GSE 510: Essential Mathematics for Economic Analysis

Total Points: 100  
10 Questions, 10 Points Each

**Instructions:** Answer all questions. Show your work clearly. Each question is worth 10 points. Partial credit will be given for correct reasoning.

## 1. Sets, Bounds, and Economic Feasibility (10 points)

Suppose a consumer can choose consumption bundles  $(x_1, x_2)$  satisfying

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \leq 10\}.$$

Answer the following:

- (a) Is  $(2, 3) \in X$ ?
- (b) Is  $(5, 1) \in X$ ?
- (c) Is  $(0, 10) \in X$ ?
- (d) For bundles in  $X$ , find the largest possible value of  $x_1$ .
- (e) For bundles in  $X$ , find the largest possible value of  $x_2$ .
- (f) Is  $X$  bounded? Briefly explain.

## 2. Logic and Proof (10 points)

Consider the statement:

$$\text{If } x > 2, \text{ then } x^2 > 4.$$

- (a) Identify the proposition  $P$  and the proposition  $Q$ .
- (b) Write the converse of the statement.
- (c) Write the contrapositive of the statement.
- (d) Is the original statement true? Briefly explain.
- (e) Is the converse true? Briefly explain.
- (f) Is the contrapositive true? Briefly explain.

## 3. Sequences, Limits, and Closed Sets (10 points)

Let

$$S = (0, 1]$$

and consider the sequence

$$x_n = \frac{1}{n}.$$

- (a) Show that  $x_n \in S$  for every positive integer  $n$ .
- (b) Find  $\lim_{n \rightarrow \infty} x_n$ .
- (c) Does the limit belong to  $S$ ?

- (d) Is  $S$  closed? Explain using your answers above.
- (e) Is  $S$  open in  $\mathbb{R}$ ? Briefly explain.

#### 4. Linear Algebra and a Market System

(10 points)

Consider a market with demand and supply:

$$Q_d = 60 - 2P, \quad Q_s = 10 + 3P.$$

In equilibrium,  $Q_d = Q_s = Q$ .

- (a) Rewrite the equilibrium conditions as a system of two linear equations in  $Q$  and  $P$ .
- (b) Write the system in matrix form  $\mathbf{Ax} = \mathbf{b}$ , where  $\mathbf{x} = (Q, P)'$ .
- (c) Compute  $|\mathbf{A}|$ .
- (d) Is  $\mathbf{A}$  nonsingular?
- (e) Solve for the equilibrium price  $P^*$  and quantity  $Q^*$ .

#### 5. Span, Linear Independence, and Interpretation

(10 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

- (a) Write a general linear combination  $a\mathbf{v}_1 + b\mathbf{v}_2$ .
- (b) Are  $\mathbf{v}_1$  and  $\mathbf{v}_2$  linearly independent?
- (c) Do  $\mathbf{v}_1$  and  $\mathbf{v}_2$  span  $\mathbb{R}^2$ ?
- (d) Can these vectors generate the vector

$$\begin{bmatrix} 4 \\ 8 \end{bmatrix}?$$

If yes, give one pair  $(a, b)$  that works.

- (e) Can these vectors generate the vector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}?$$

Explain.

- (f) Give a brief economic interpretation of what it means for two activity vectors not to span  $\mathbb{R}^2$ .

#### 6. OLS in Matrix Form

(10 points)

Suppose we estimate the regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

using the data

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

- (a) Write the model in matrix form.
- (b) Compute  $\mathbf{X}'\mathbf{X}$ .
- (c) Compute  $\mathbf{X}'\mathbf{y}$ .
- (d) Compute  $(\mathbf{X}'\mathbf{X})^{-1}$ .

(e) Compute

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

(f) Write the fitted regression line.

### 7. Differentiation and Optimization

(10 points)

Suppose a firm has revenue and cost

$$R(q) = 80q - 2q^2, \quad C(q) = 20 + 10q + q^2.$$

Profit is

$$\pi(q) = R(q) - C(q).$$

- (a) Write profit  $\pi(q)$  as a function of  $q$ .
- (b) Find  $\pi'(q)$ .
- (c) Solve the first-order condition  $\pi'(q) = 0$ .
- (d) Find  $\pi''(q)$ .
- (e) Is the critical point a local maximum or a local minimum?
- (f) Find the maximum profit.

### 8. Linear and Quadratic Approximation

(10 points)

Suppose output is produced according to

$$Y = F(K) = 20K^{1/2}.$$

The current capital stock is  $K_0 = 100$ .

- (a) Find  $F'(K)$  and  $F''(K)$ .
- (b) Compute  $F(100)$ .
- (c) Use the linear approximation around  $K_0 = 100$  to approximate  $F(109)$ .
- (d) Use the quadratic approximation

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2$$

to approximate  $F(121)$ .

- (e) Compute the exact value  $F(121)$ .
- (f) Which approximation is closer to the exact value?

### 9. Implicit Differentiation and Indifference Curves

(10 points)

A consumer's indifference curve is described implicitly by

$$F(x, y) = x^{1/2} + y^{1/2} - 5 = 0.$$

- (a) Compute  $F_x$  and  $F_y$ .
- (b) Use the implicit function theorem formula

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

to find the slope of the indifference curve.

- (c) Evaluate  $\frac{dy}{dx}$  at  $(x, y) = (4, 9)$ .
- (d) Use a linear approximation to estimate how much  $y$  must change if  $x$  increases from 4 to 4.2 while remaining on the same indifference curve.
- (e) Explain why the sign of your answer makes economic sense.

# 10. Comparative Statics and the Implicit Function Theorem

(10 points)

Suppose a firm chooses output  $q$  to maximize

$$\pi(q, t) = 100q - \frac{1}{2}q^2 - tq,$$

where  $t$  is a per-unit tax.

- (a) Find the first-order condition for profit maximization.
- (b) Solve explicitly for optimal output  $q^*(t)$ .
- (c) Find  $\frac{dq^*}{dt}$  using your explicit solution.
- (d) Now define

$$F(q, t) = 100 - q - t.$$

Use the implicit function theorem formula

$$\frac{dq^*}{dt} = -\frac{F_t}{F_q}$$

to verify your answer from part (c).

- (e) Suppose the current tax is  $t = 20$ . Find  $q^*(20)$ .
- (f) Use a linear approximation to estimate optimal output if the tax increases from 20 to 23.
- (g) Interpret the sign of  $\frac{dq^*}{dt}$  economically.